

Section A

Answer **all** the questions in this section.

- 1 **Fig. 1.1** shows the sequence of bases in a section of a gene coding for the first seven amino acids of a polypeptide chain.

5' AA ACTGTTTCTGCTGGAACAT 3'

Fig. 1.1

- (a) (i) Methionine is coded by the initiation codon. On **Fig. 1.1**, label the 3' and 5' end of the DNA template strand. [1]
- (ii) List the corresponding codons in the mRNA transcribed from the template in **Fig. 1.1**.

AUG UUC CAG CAG AAA CAG UUU
A: 3' UUU GAC AAA GAC GAC CUU GUA 5'

[1]

The polypeptide was hydrolysed to release the first seven amino acids. It contained four different amino acids. The number of each type obtained is shown in the **Table 1.1**.

Amino acid	Number present
Phe	2
Met	1
Lys	1
Gln	3

Table 1.1

- (iii) Using information in **Table 1.1**, work out the order of amino acids in the polypeptide.

met phe gln gln lys gln phe

R: reverse sequence

[1]

- (b) (i) Explain why a single base deletion in the DNA sequence is more likely to produce non-functional proteins than a single base substitution.

- 1. deletion results in frameshift**
- 2. Change in sequence of codons thus /causing all amino acids downstream of mutation to change**
- 3. substitution usually cause change in 1 aa/1 codon;**
- 4. or may result in silent mutation that results in no change in aa (if it occurs at 3rd base as a result of degeneracy of genetic code) Or aa has similar property**
- 5. Substitution may not lead to conformational change of protein but deletion mutation will. WTTE**

[3]

(ii) Describe the effect of replacing guanine (underlined in **Fig.1.1**) with adenine.

- **AAG → AAA / UUC → UUU**
 - **Due to degenerate code (as more than one codon may code for the same aa);**
 - **still code for phe / no change in aa (silent mutation);**
- [2]

(c) Explain how the structure of the nuclear envelope facilitates translation.

1. The presence of **nuclear pores**;
2. Which provide the passage for mature **mRNA** to **move out** of nucleus to ribosome in cytoplasm for translation;
3. **Ribosomal subunits** to **move out** of nucleus to cytoplasm to form ribosome / **tRNA** **move out** of nucleus to cytoplasm to undergo amino acid activation;
4. **Ribosomal proteins** to **move into** nucleus from cytoplasm to form ribosomal subunit;

Pt 1 must be in

Mark for 2 to 4 is awarded for direction of movement and molecule

[2]

(d) A variety of related proteins can be produced from a single gene. State the process and explain how the same gene can produce different protein products.

1. **Alternative splicing**;
(Spliceosome binds to intron-exon boundaries in the pre-mRNA)
2. Resulting in **different exons being spliced together** in different ways leading to formation of many different mature mRNA resulting in different proteins translated

[2]

[Total : 12]

2 Chromosome aberrations can have devastating effects on human health.

- (a) Down syndrome is a disorder caused by an aberration in chromosomal number. Individuals with Down syndrome have three copies of chromosome 21, which typically results in some delay in cognitive ability and physical growth, as well as a characteristic set of facial features.

Fig. 2.1 is a diagrammatic representation of chromosome 21, showing the location of two different RFLP loci.

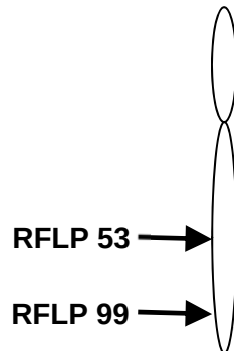


Fig. 2.1

Fig. 2.2 is a pedigree of a family and the results following restriction digestion, gel electrophoresis and Southern blotting with specific probes of the genetic material of some family members.

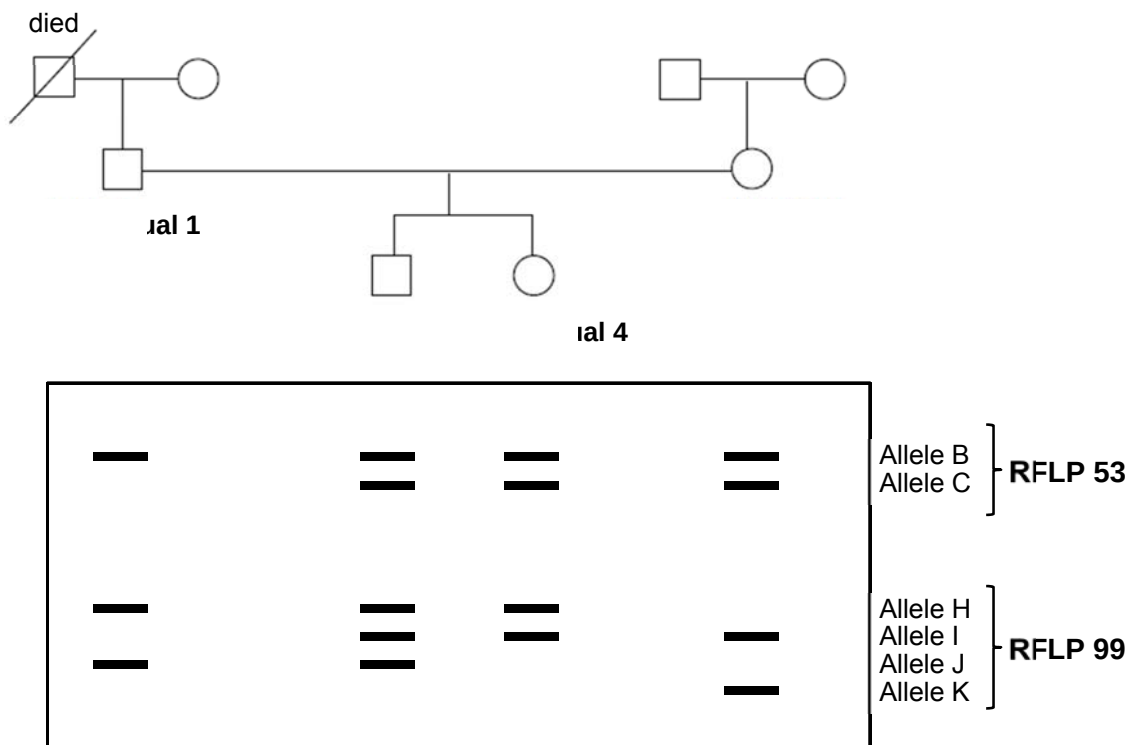


Fig. 2.2

- (i) Identify which individual (3 or 4) has three copies of chromosome 21. Give a reason for your answer.

1. Individual 3

Reasons (any one):

2. 3 bands/alleles of RFLP 99

Or

3. Has 1 copy of allele H, 1 copy of allele I and 1 copy of allele J of RFLP 99

[2]

- (ii) Non-disjunction of chromosomes during the formation of gametes would have resulted in gametes with two copies of chromosome 21.

In which parent (individual 1 or 2) did the non-disjunction of chromosomes occur?

1. Individual 1 [1]

- (iii) State when the non-disjunction event occurred. Explain your answer.

(Assuming that there is no crossing-over. This is a valid assumption as there is no mention of crossing over occurring in the question. This was the assumption that you were expected to make.)

1. Anaphase I or Meiosis I

2. Homologous chromosomes with different combinations of RFLP alleles (BH on one and BJ on another) move to the same daughter cell

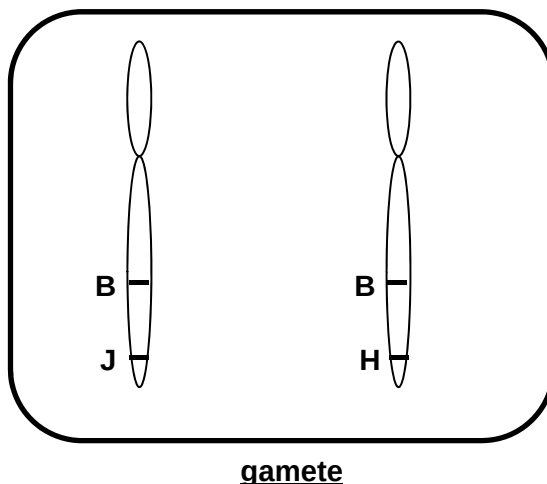
[2]

(However if crossing over occurred in Prophase I between RFLP 53 and RFLP 99.)

3. Anaphase II or Meiosis II

4. Crossing over occurred between RFLP 53 and RFLP 99 (in prophase I) and the non-identical sister chromatids moved to the same daughter cell in Meiosis II

- (iv) Draw chromosome(s) 21 in the gamete that resulted in the individual having Down syndrome. Clearly label the alleles at the RFLP loci on each homologue of chromosome 21.



- (b) Fig. 2.3 shows the formation of a Philadelphia chromosome which is a chromosomal abnormality that results from a reciprocal translocation between chromosome 9 and 22. This chromosomal abnormality is associated with chronic myelogenous leukemia which is considered a form of cancer. [1]

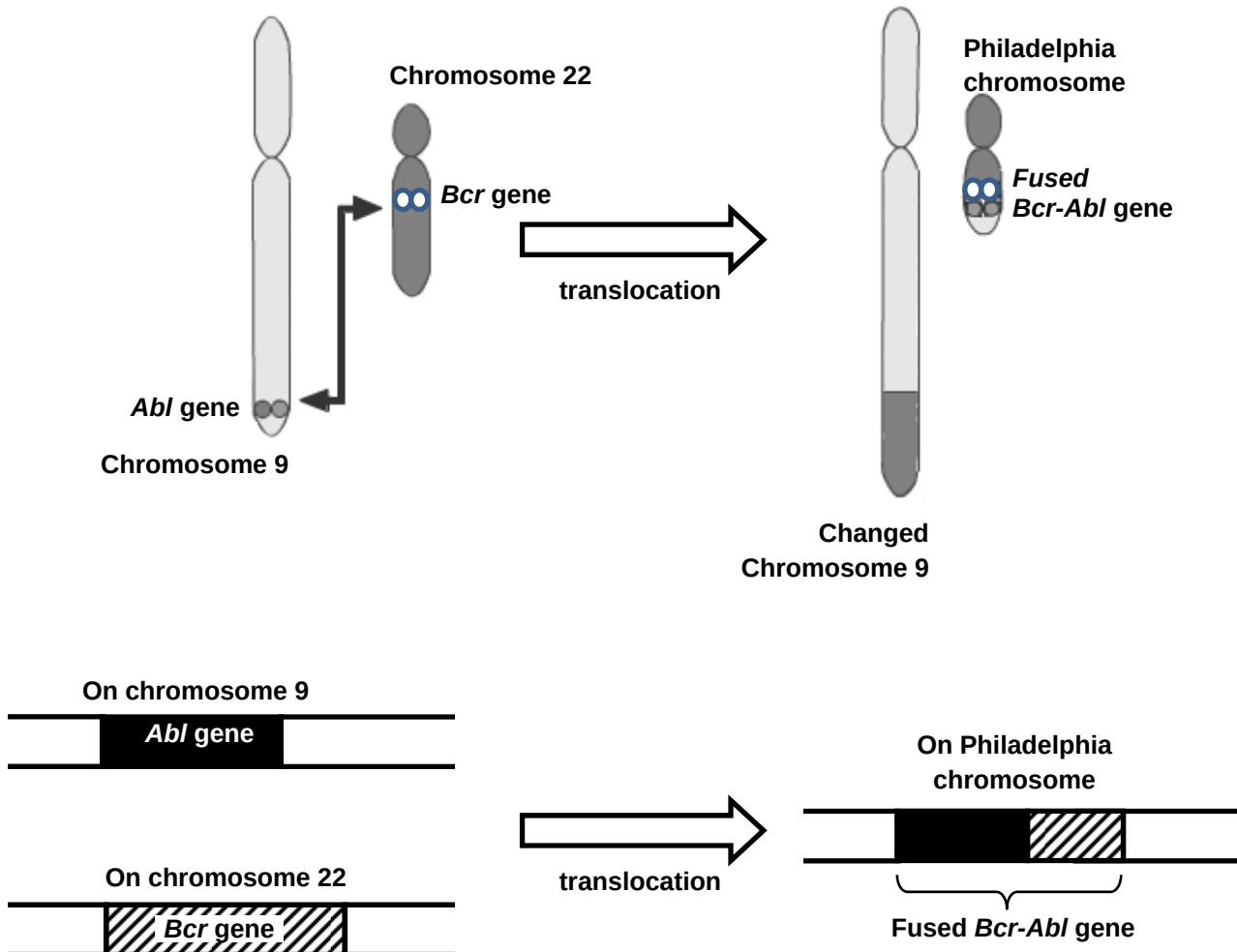


Fig. 2.3

The *Abl* gene normally codes for a receptor involved in **signal transduction** that results in the cell proliferation.

Using the information in Fig. 2.3, suggest how the fused gene product can result in cancer.

1. Fused gene is an **oncogene***
2. As a result of **gain-in-function*** mutation of the gene
3. Which results in the formation of abnormal/different receptor that is able to activate relay proteins **without signal molecules** (wtte)
or
Constitutively active (even without signal)
4. Which causes **uncontrolled cell division/proliferation**

[3]

(c) Cancer cells require tetrahydrofolate to make purines and pyrimidines for cell proliferation. Conversion of precursors to tetrahydrofolate is carried out by the enzyme dihydrofolate reductase (DHFR). Competitive inhibition of DHFR by administering the drug Methotrexate (MTX) is a key treatment for certain cancers such as acute lymphoblastic leukemia.

(i) Using the information provided, comment on the structure of Methotrexate.

1. Methotrexate is a competitive inhibitor and is complementary in shape and charge to the active site of DHFR.

Or

Similar in shape and charge to the precursors of tetrahydrofolate

[1]

(ii) However, researchers have since found that one of the main causes of failure in the treatment of acute lymphoblastic leukemia is MTX resistance due to the amplification of the DHFR gene.

During the early stages of the research, scientists suggested a mechanism for the amplification of the DHFR gene as shown in **Fig. 2.4**.

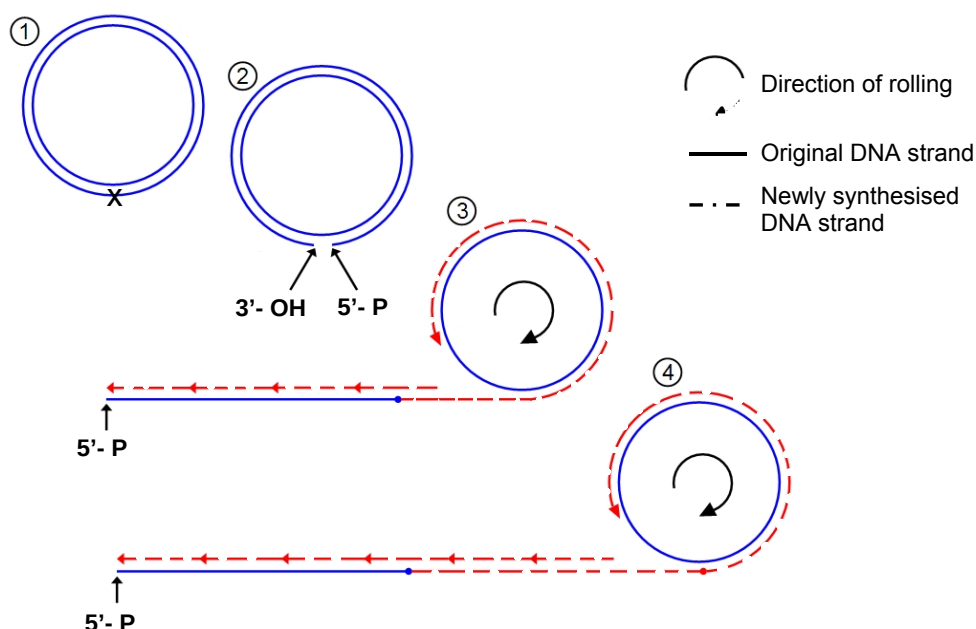


Fig. 2.4

With reference to **Fig. 2.4**, describe the process.

1. DHFR gene is excised and circularised;
2. (1) One strand of a double-stranded DNA is nicked/cut at a site;
3. (2) Using the un-nicked/intact strand as a template;
4. (DNA polymerase) adds nucleotides to the free 3'-OH end to synthesise a complementary DNA strand;
5. (3) 5'-P end of the nicked strand is displaced as synthesis/strand elongation proceeds;
6. (3) + (4) Displaced strand can be used as a template and replicated discontinuously;

[3]

[Total : 13]

- 3(a)** A study was conducted on the *lac* operon of *E. coli*. **Fig. 3.1** below shows the activity of β -galactosidase as a function of time for a culture of *E. coli*. The bacterial cells were initially grown in glycerol as the only carbon source. At time A, lactose was added to the culture. When the lactose supply is exhausted at time B, glucose was added.

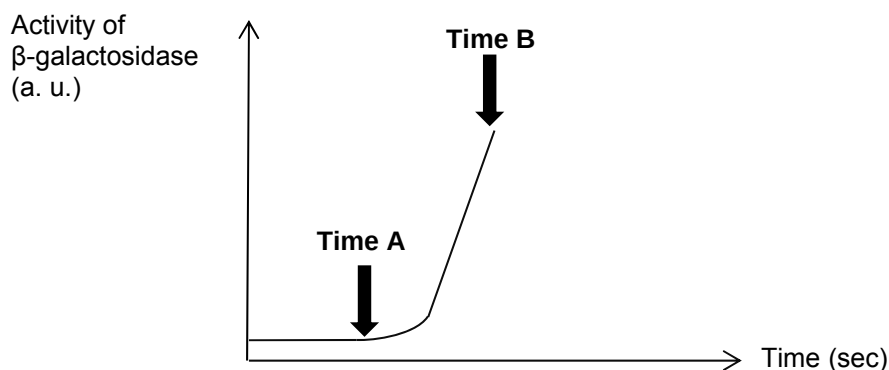


Fig. 3.1

- (i)** Why is the *lac* operon considered inducible?

1. Expression of the 3 genes (*lac Z*, *lac Y* and *lac A**) is usually “off”/not transcribe ;
2. but can be turned “on”/transcribed in the presence of inducer allolactose/lactose ;

[2]

- (ii)** Explain the change in β -galactosidase activity from time A to B. [4]

High lactose,

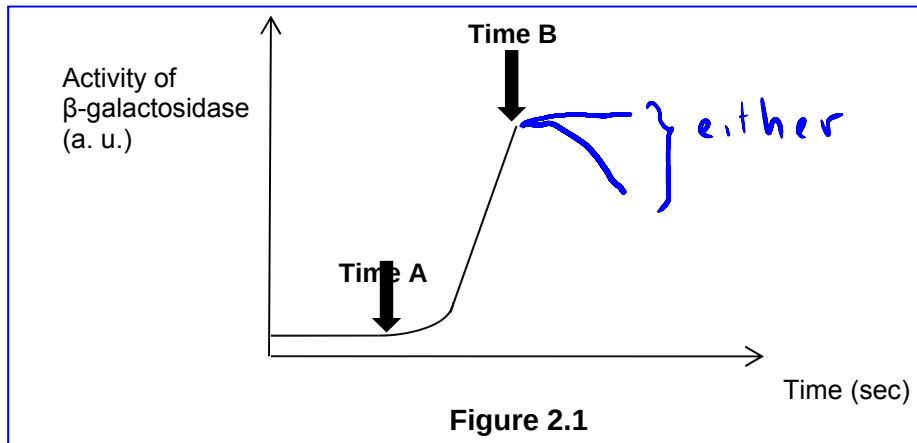
1. Time is needed for the expression /activation of *lac* operon to produce enzyme before lactose can be metabolised
2. Lactose converted to allolactose
3. Lactose/ allolactose binds to and inactivates/alters tertiary structure of repressor such that it fails to bind to operator
4. RNA polymerase is free to bind to promoter and actively transcribe structural genes of the operon.

Low glucose

5. High **cAMP*** will result in active catabolite activator protein (CAP) that strengthen affinity of promoter for RNA polymerase. Thus, rate of transcription increases

[4]

- (iii) Draw in **Fig. 3.1** what a continuation of the graph would look like from time B. [1]



- (b) **Fig. 3.2** shows the arrangement of the *lac* operon and its regulatory gene.

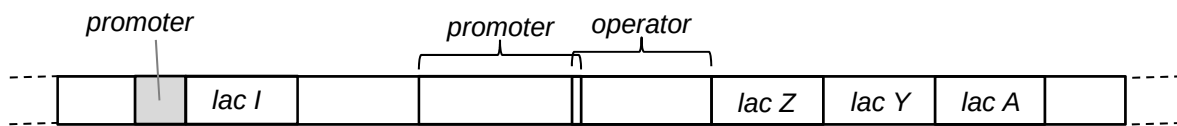


Fig. 3.2

A series of mutations was introduced to the *lac* operon.

- (i) The operator is deleted. Assuming lactose is present, what is the effect of this on the transcription of *lac Z*, *lac Y* and *lac A* genes?

1. There is no operator for the repressor to bind to thus;
2. RNA polymerase can bind to promoter to start transcription of the genes;

[2]

- (ii) A point mutation was introduced into *lac Z*, resulting in the production of a truncated non-functional protein. Assuming lactose is present, explain if the proteins encoded by *lac Y* and *lac A* genes are produced?

1. Yes they will be produced;
2. The enzymes are translated separately because each has its own start and stop codon on the polycistronic mRNA, so an unexpected stop codon in one does not affect the others;

[2]

- (c) An *E. coli* strain was isolated from the intestines of a mammal. The bacterium was found to have two different copies of the *lac* operon. Describe a genetic transfer process that resulted in this observation.

1. **Conjugation*** - F plasmid carrying a second copy of *lac* operon is transferred to this bacterial cell;
2. **Transduction*** - bacteriophage added a second copy of *lac* operon from a previous host;
3. **Transformation*** - DNA fragment containing a second copy of *lac* operon taken up by cell;

[1]

[Total : 12]

- 4 An **absorption** spectrum is a graph of the absorption of different wavelengths of light by a photosynthetic pigment.

An **action** spectrum is a graph of the rate of photosynthesis at different wavelengths of light.

Fig. 4.1 shows the absorption spectra of chlorophyll a and chlorophyll b as well as an action spectrum.

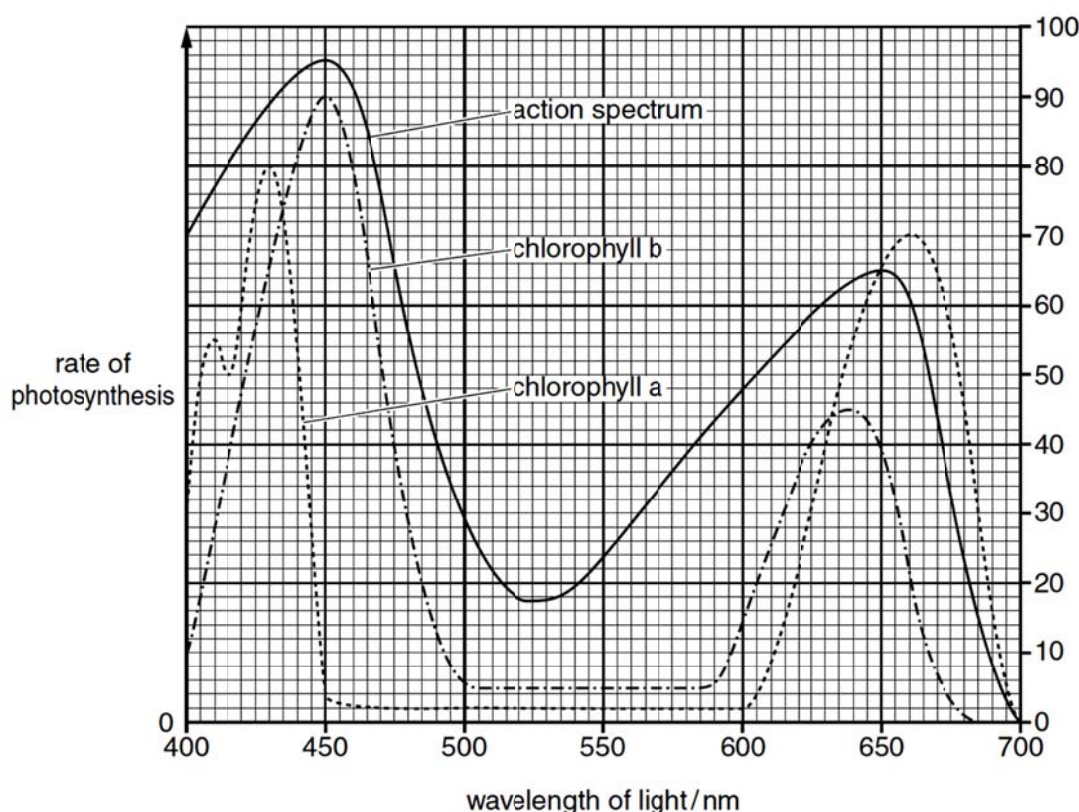


Fig. 4.1

- (a) With reference to **Fig. 4.1**,
 (i) compare the **absorption** spectra of chlorophyll a and chlorophyll b.

Differences

1. For percentage of light absorption, chlorophyll a peaks at 430nm with 80% of light absorption while chlorophyll b peaks at 450nm with a higher 90% light absorption;
2. Chlorophyll a peaks at 660nm with much higher light absorption of 70% while chlorophyll b peaks at 635–640nm with 45% light absorption/ peak of light absorption for chlorophyll a at 660nm.

Similarities

3. Both shows higher light absorption of blue light (400–500nm) and red light (600–700nm) than other wavelengths;
4. Both peaks in blue light are higher than red light;
5. Both have little/least absorption, between 500–600nm / in green light;

Or

Little light absorption for chlorophyll a between 450–600nm and for chlorophyll b between 500–600nm ;

Mark awarded as follows:

Ref. wavelength

1 chlorophyll a peaks at 430nm and chlorophyll b peaks at 450nm ;

Or

2 chlorophyll a peaks at 660nm and chlorophyll b peaks at 635–640nm ;

3 ref. linking 400–500nm with blue light and ref. linking 600–700nm with red light ;

4 both have little absorption, between 500–590/600nm / in green light ;

A: little absorption, chlorophyll a 450–600 and chlorophyll b 500–600 ;

Ref. light absorption

5 both peaks in blue light are higher than peaks in red light ;

6 chlorophyll b higher than chlorophyll a in the blue light / chlorophyll a higher than chlorophyll b in the red light / AW ; A converse

7 comparative figures on % of light absorption for pt 1(90% vs 80%) or 2(70% vs 45%) and 5(eg. for b 90% in blue light vs 45% in red light) and 6(e.g. peak of b is 10% higher than a for blue light)

Nb: only awarded if specific wavelength is quoted ;

[2]

(ii) explain the shape of the **action** spectrum.

1. Peak(s) of action spectrum/ highest rate of photosynthesis, correspond to absorption peak(s);
2. Light absorbed by different photosynthetic pigments is used for photoactivation/photosynthesis thus the higher percentage of light absorption leads to higher rate of photosynthesis in red and blue light ;
3. The region between 500 to 600nm is not an exact match between absorption and action spectra
4. due to the role of carotenoids and other accessory pigments in light absorption that pass the energy via resonance to the reaction centre
A: chlorophyll a (link to pt 3) [2]

(b) The light-dependent stage of photosynthesis takes place on the thylakoids of the chloroplast. **Fig. 4.2** represents a section through a thylakoid to show some of the components involved in the light-dependent stage.

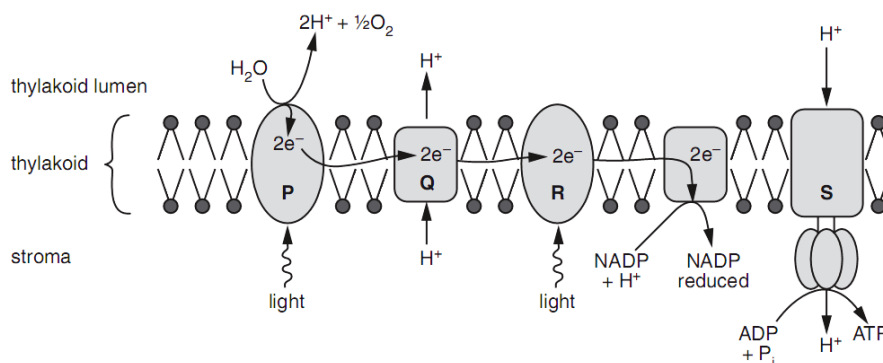


Fig. 4.2

- (i) Identify the structures labelled R and S.

R : Photosystem I/Photosystem 700 S : ATP synthase/ stalked particle [2]

- (ii) Atrazine is a widely used herbicide (weed killer). It binds to a chloroplast protein involved in electron transfer between photosystems.

Explain how atrazine works as an effective herbicide.

1. Electron cannot be passed down the electron transport chain* A: electron carriers (ETC);
2. No reduced NADP/ NADPH + H⁺ produced (linked to 1);
3. Resulting in no / reduced, pumping of protons across the thylakoid membrane from stroma to thylakoid lumen and thus no proton gradient / proton motive force
4. Chemiosmosis does not occur thus no ATP is produced (linked to 3)
5. no carbon fixation / Calvin cycle / light independent stage thus no glucose/triose phosphate(as glycerate phosphate cannot be converted to glyceraldehyde-3-phosphate/triose phosphate) formed for respiration and plant dies ; [3]

- (c) In an investigation, mammalian liver cells were homogenised (broken up) and the resulting homogenate centrifuged. Samples of the complete homogenate and samples with only mitochondria were incubated with:

- 1 glucose
- 2 pyruvate
- 3 glucose and cyanide
- 4 pyruvate and cyanide

Cyanide inhibits oxidative phosphorylation.

After incubation, the presence or absence of carbon dioxide and lactate in each sample was determined. The results are summarised in **Table 4.1**.

	Sample of homogenate			
	Complete		Only mitochondria	
	Carbon dioxide	Lactate	Carbon dioxide	Lactate
1 glucose	✓	✓	X	X
2 pyruvate	✓	✓	✓	X
3 glucose and cyanide	X	✓	X	X
4 pyruvate and cyanide	X	✓	X	X

X = absent ✓ = present

Table 4.1

- (i) Explain why carbon dioxide is produced when mitochondria are incubated with pyruvate but **not** when they are incubated with glucose.

1. Pyruvate can enter the mitochondrion but glucose cannot
2. Glucose needs to be converted to pyruvate via glycolysis that takes place only in the cytoplasm/ cytosol;
3. Carbon dioxide is produced by decarboxylation* of pyruvate via link reaction / of acetyl CoA in Krebs cycle in mitochondrion; [2]

- (ii) With reference to **Table 4.1**, explain the difference in the results when the complete cell homogenate is incubated with **2** and **4**.

1. Carbon dioxide is produced only when incubated in 2;
2. oxidative phosphorylation cannot take place and reduced NAD is not oxidised/ NAD is not regenerated AW ;
3. Krebs cycle /link reaction stops so no decarboxylation takes place ; [2]

[Total : 13]

- 5 Grove snails are known to have brightly coloured shells, with the alleles C^P and C^Y coding for pink and yellow coloured shells respectively. Shells of the grove snails have different banding patterns as well, with allele B^B conferring a shell with bands and allele B^O conferring an unbanded shell.



Banded snail

Unbanded



snail

A grove snail with a banded pink shell is crossed with another grove snail with an unbanded yellow shell, and the resultant F_1 generation all had unbanded pink shells. The F_1 were test-crossed and gave rise to 250 offspring with the following phenotypes:

banded pink shell	121
unbanded pink shell	14
banded yellow shell	11
unbanded yellow shell	104

- (a) State the expected offspring phenotypic ratio of the test cross.

1 banded pink shell : 1 unbanded pink shell : 1 banded yellow shell : 1 unbanded yellow shell

..... [1]

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

$$\chi^2 \text{ test: } \chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

Key to symbols: Σ = sum of ...
 v = degrees of freedom
 c = number of classes
 O = observed value
 E = expected value

- (b) (i) Showing your working clearly, calculate the χ^2 value in the space below.

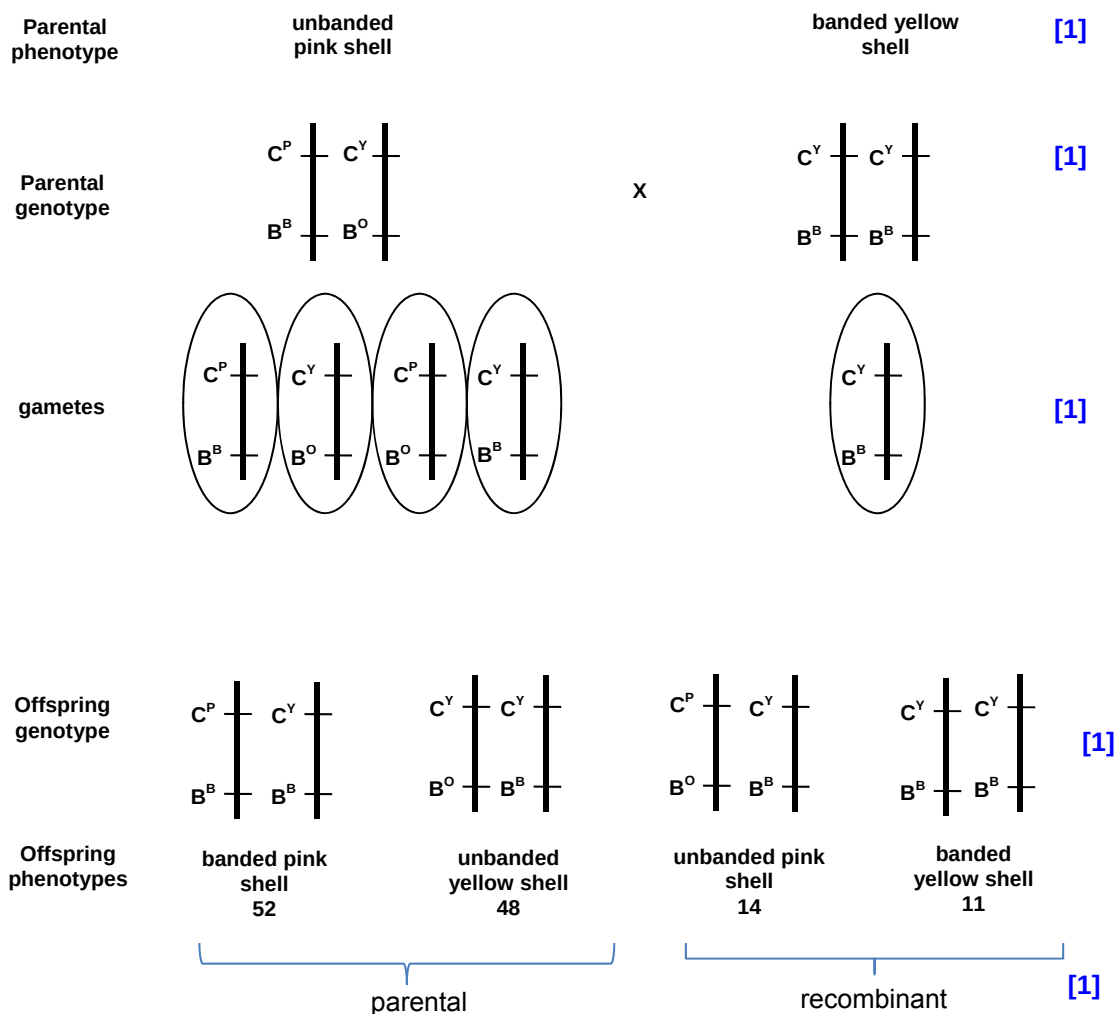
$$\begin{aligned} \chi^2 &= \frac{(121 - 62.5)^2}{62.5} + \frac{(104 - 62.5)^2}{62.5} + \frac{(14 - 62.5)^2}{62.5} + \frac{(11 - 62.5)^2}{62.5} \\ &= 54.756 + 27.556 + 37.636 + 42.436 \\ &= 162.384 \end{aligned}$$

1 Mark for working 1 Mark for result

- (ii) Using the χ^2 table, explain the conclusion drawn from the calculated χ^2 value in b(i). [2]

1. df = 3 and p < 0.001
2. At 5% significant level (if p < 0.001 mentioned)/Since p < 0.05, difference between expected and observed results is significant and is due to some other factor and not by chance alone.
3. May be due to autosomal linkage [2]

- (c) With the use of a genetic diagram, explain the observed results of the test cross.



[5]

The same test cross was carried out another time and the resulting number of banded pink snails and unbanded yellow snails were equal. The resulting number of unbanded pink snails was also equal to that of the banded yellow snails. Through this test cross, the 2 gene loci are determined to be 28 map units apart.

- (d) State the observed phenotypic ratio of the offspring from this test cross.

18 banded pink shell : 18 unbanded yellow shell : 7 unbanded pink shell : 7 banded yellow shell [1]

[Total : 11]

- 6 Homeostatic mechanisms maintain a constant environment in the body.
Fig. 6.1 shows changes in plasma glucose concentration that occurred in a person who went without food for some time.

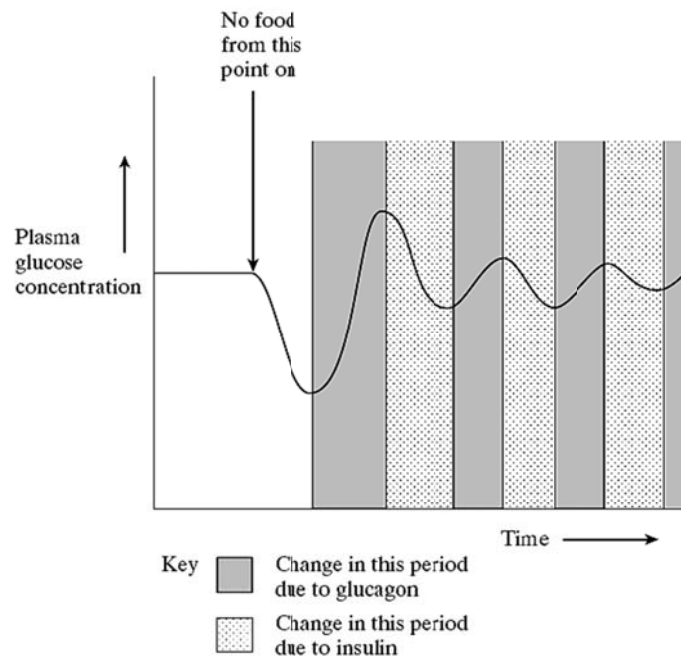


Fig. 6.1

(a) Use evidence from Fig. 6.1 to explain the change in plasma glucose concentration.

1. No food given thus decrease in plasma/blood glucose concentration below the norm level/set point of 90mg/100ml is ;
2. detected by the α cells of islet of Langerhans in pancreas which stimulates the (increased) secretion of glucagon into the blood stream;
3. Glucagon binds to cell surface receptors of the effector e.g. liver cell that stimulates the conversion of glycogen to glucose/gluconeogenesis/conversion of amino acids to glucose increase in blood glucose via negative feedback; R: if muscle cell mentioned
4. detected by β cells in pancreas that stimulate (increased) secretion of insulin into the blood;
5. insulin is transported to the effector e.g. muscle/liver cell that results in increase uptake of glucose/membrane permeability to glucose that reduces the blood glucose level via negative feedback* (can be in pt 3);
6. Antagonistic action of both hormones result in fluctuations of blood glucose concentration about the norm which diminished over time, eventually reaching norm;

[4]

Adrenaline is a signal molecule that stimulates similar cellular responses from liver cells and adipose cells in a way similar to that of glucagon.

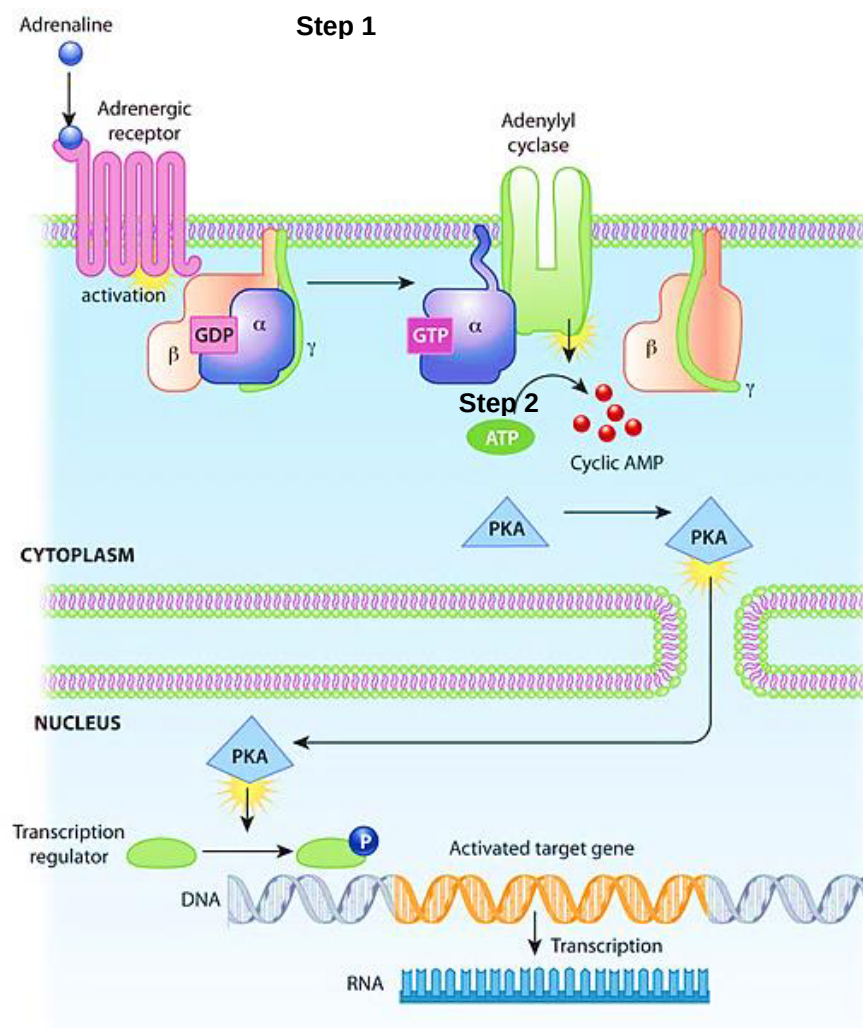


Fig. 6.2

(b) With reference to Fig. 6.2,

(i) describe the events occurring in step 1 after the binding of adrenaline molecule to the adrenergic receptor.

1. Binding results in a conformational change of the receptor at the cytoplasmic region/intracellular domain of the receptor ;
2. That allows binding to G protein that activates it by replacing GDP with GTP ;
3. α subunit dissociates from the β and γ subunits, moves along the plasma membrane and binds with and activates the adenylyl cyclase ;

[2]

(ii) explain the significance of step 2 that results in effective signal transduction.

1. An activated adenylyl cyclase catalyses the formation of many cAMP from many ATP
2. Thus amplifying the signal as
3. Idea of a single ligand/ligand can produce a large cellular response (Any 2 of 1-3)
Or
4. cAMP is a second messenger* that is small and diffusible and

5. can diffuse easily in the cell to activate the next relay protein (2M for 4&5)

(iii) describe the series of events that occur after the activation of protein kinase A that results in a cellular response. [2]

1. Activated protein kinase A passes through the nuclear pore into the nucleus ;
2. Phosphorylate transcription factor resulting in a conformation change/activation thus allowing it to
3. bind to a specific regulatory DNA sequence e.g. enhancer
4. recruiting RNA polymerase or promote assembly of transcription initiation complex/ result in the transcription of the target gene ;

[2]

[Total : 10]

7 For a long time, the prevailing idea among evolutionary biologists is that geographical barriers are a prerequisite for speciation. However, recently, more and more examples of an alternative speciation process which does not require a geographical barrier were discovered but this remains the exception rather than a norm.

(a) (i) What alternative speciation process are we referring to?

Sympatric speciation*

[1]

One case study involved studying the palm genus *Howea* from Lord Howe Island, a tiny volcanic island in the Tasman Sea. The two related palm species found on the island have a different soil preference. The different types of soil are interspersed throughout the island and pollen from the palms is wind-dispersed.

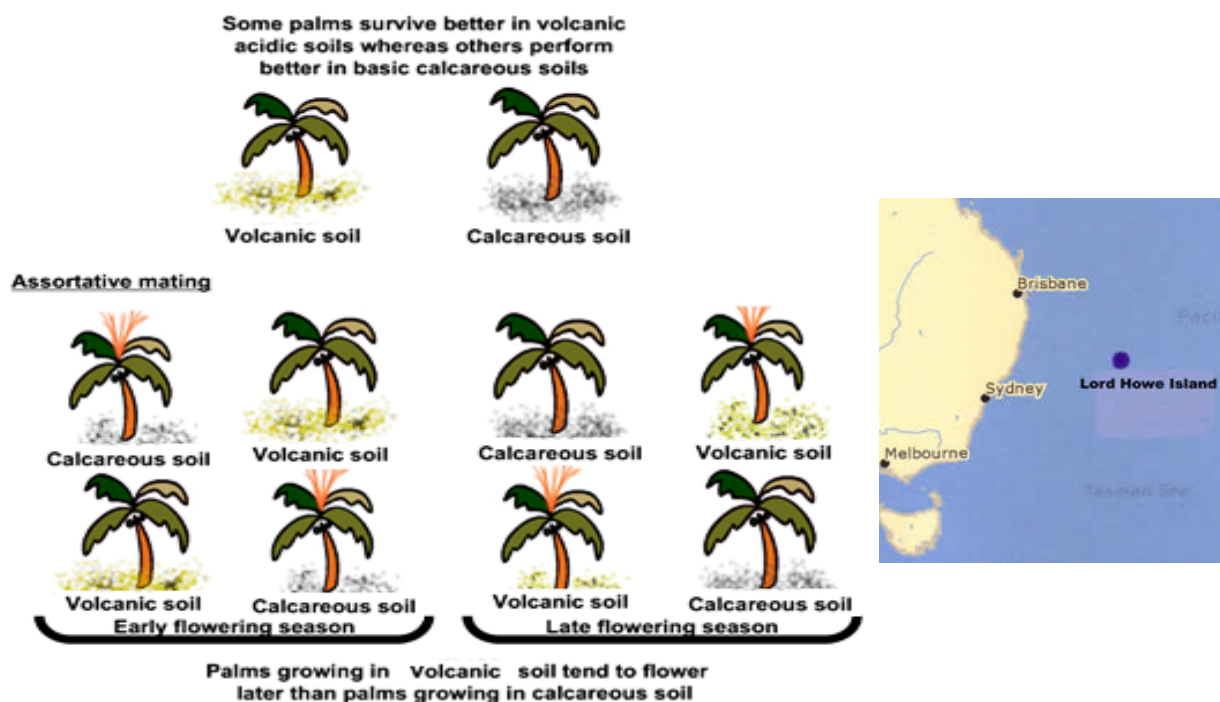


Fig. 7.1

(ii) Explain how speciation could have taken place to give rise to the two sister species from a common ancestor.

1. Ancestral palm growing on acidic volcanic soil developed a later/different flowering time compared to palms growing on basic calcareous soil that flowered earlier;
2. resulting in reproductive/physiological isolation*;
3. This prevented interbreeding and disrupted the gene flow* between the two populations of palm;
4. As the different populations evolve independently from each other, they accumulate different genetic mutations* over time;
5. With variation among the palms and exposure to different selection pressures, natural selection will select for those best adapted to their environment,
6. who are more likely to survive, reproduce and passed on their alleles to the next generation;
7. Over time, allele frequencies change can also be due to genetic drift

[5]

(iii) Give an evidence why scientists can rule out geographical isolation mechanisms?

It is a tiny island;

R: no major barrier only

Pollen grains can be easily dispersed by wind;

Different soil where the 2 species grow on are interspersed throughout the island;

[1]

- (b) *Howea belmoreana* goes by the names, curly palm, kentia palm and Belmore sentry palm while *Howea forsteriana* goes by the names thatch palm and kentia palm.

Explain why the scientific name is better than the common names.

1. It is unambiguous and precise with one unique name for one species;
2. unlike the common name "Kentia palm" which is used to refer to both species or multiple names referring to one species (evidence);
Or
3. The binomial nomenclature is a systematic naming system that allows one to categorise/group similar organisms together involving both the species name and genus name (evidence);
4. while the common names do not reveal their position in the plant kingdom nor attempts to categorise them; ORA

1 and 2 (link) or 3 and 4 (link). If 1 and 3 or 4 allow only 1 mark (unlink)

[2]

[Total: 9]

Section B Answer EITHER 8 OR 9.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 8 (a) Explain how meiosis can contribute to genetic variation in offspring. [6]

1. Meiosis produces haploid daughter cells/gametes that are genetically different from the parent, contributing to genetic variation;
2. During prophase I*, crossing over* of corresponding segments of non-sister* chromatids of homologous chromosomes/bivalent at chiasmata;
3. leads to exchange of genetic materials / linkage groups are broken to form non-identical sister chromatids with new combinations of alleles on chromosomes;
4. During metaphase I*, independent assortment of homologous chromosomes* occurs when arrangement of one pair of homologues at the metaphase plate is independent of the arrangement of the other pairs of homologues;
5. New combination of maternal and paternal chromosomes;
6. Subsequently chromosomes of each homologous pair separate independently during anaphase I*;
7. Random/independent arrangement of non-identical sister chromatids at the metaphase plate during metaphase II and subsequent separation of non-identical sister chromatids during anaphase II;
8. This results in gametes with new combinations of alleles that differ from parental combination of alleles with 2^n * possible combinations/types of gametes where n is number of homologous pairs of chromosomes; (1m as long as "new combination of allele" has been written)

9. During fertilization*, random fusion* of gametes;
10. results in greater number of genotypic combinations of a zygote;

- (b) Discuss how bacteria evolve through natural selection and the implication in antibiotic resistance. [8]

how antibiotic resistance in bacteria arises:

1. Due to spontaneous mutation, there is existing variation* in the population of bacteria with antibiotic resistant strains and non-resistant strains;
 2. (idea of horizontal gene transfer) The allele for antibiotic resistance can also be acquired through conjugation* / transduction* / transformation* (1m = need to state the process + elaborate process) [2m max]
 3. When exposed to antibiotic, antibiotic act as selection pressure* to select against/kill those non-resistant bacteria;
 4. while those bacteria with antibiotic resistance allele are selected for/best adapted and survive;
 5. and pass on the allele for antibiotic resistance to subsequent generations of bacteria cells during binary fission thus most of the bacteria are resistant to antibiotic (idea of vertical gene transfer);
 6. allele frequency changes as a result of natural selection;
- implication in antibiotic resistance:
7. Over many generations, bacteria cells evolve by natural selection and the bacteria become resistant to the antibiotic rendering that antibiotic ineffective;
 8. Thus, other types / different combinations of antibiotics have to be used;
 9. possibility of developing multiple antibiotic resistance;

- (c) Describe the link between the frequency of sickle cell anaemia allele and the number of cases of malaria. [6]

1. Frequency of sickle cell anaemia allele is highest in regions of endemic malaria;
2. Individuals with two recessive Hb^S alleles/ $Hb^S Hb^S$ develop sickle cell disease as sickle red blood cells exhibit poor oxygen transport;
3. Homozygous individuals with two normal Hb^A alleles/ $Hb^A Hb^A$ have normal red blood cells;
4. but have a greater risk of dying from malaria in regions of endemic malaria;
5. Heterozygous individuals with one recessive Hb^S allele/ $Hb^A Hb^S$ produce both normal and abnormal haemoglobin; (idea of carrier with sickle cell trait)

Heterozygote protection:

6. Idea of heterozygote protection plus type of selection pressure: Heterozygous condition hide recessive Hb^S allele that is less favourable from natural selection which only acts on sickle cell anaemia phenotypes
7. thus maintaining the recessive allele in the population;

Heterozygote advantage:

8. Heterozygotes are less likely to suffer from malaria thus they have selective advantage over homozygotes/heterozygote advantage;
9. Sickle shape of red blood cell results in lower oxygen carrying capacity thus malaria parasite cannot survive;
Or idea that malaria parasite cannot survive well in cell with Hb^S
10. Heterozygotes are more likely to survive and reproduce to pass on the Hb^S allele;

[Total: 20]

- 9 (a) Explain how gene expression can be controlled at the translational level and describe the mechanisms present in prokaryotes and eukaryotes. [8]

	Prokaryotes	Eukaryotes
mRNA half-life	1. Both involve regulation of mRNA half-life → longer half-life allows for translation of more proteins from mRNA; 2. mRNA rapidly degraded / has short half life to allow for fast changes in gene expression; 3. Binding of <u>anti-sense mRNA</u> can target mRNA for degradation, further decreasing its half-life;	4. Half-life of mRNA influenced by <u>length of 3' poly-A tail</u> ; Or when the polyA tail shorten to a critical length, 5' cap is removed and mRNA is degraded 5. longer 3' poly-A tail = more stable / longer half life; 6. siRNA bind to target mRNA for degradation
Binding to small ribosomal subunit	7. Translation can be blocked by <u>preventing the binding of the small ribosomal subunit to the mRNA</u> for assembly of the translation initiation complex; 8. Small ribosomal subunit recognises and binds to <u>Shine Dalgarno region</u> on mRNA for <u>proper positioning of start codon</u> ; 9. <u>Translational repressor protein</u> or <u>anti-sense mRNA</u> binds to / near <u>Shine Dalgarno region</u> on mRNA to prevent binding of small ribosomal subunit;	10. Interaction between 5' cap and 3' poly-A tail is required for translation initiation; 11. <u>Translational repressor protein</u> binds to <u>5' cap</u> or <u>3' poly-A tail</u> to prevent interaction;
Availability of initiation factors	12. <u>Initiation factors</u> required for proper positioning of the small ribosomal subunit together with initiator tRNA on the mRNA, and recruitment of the large ribosomal subunit / assembly of translation initiation complex;	

- (b) Explain how the structure of specific transcription factors is linked to their roles in control of eukaryotic gene expression. [7]

1. Specific transcription factors: activators & repressors ;
2. 2 regions/domains: DNA binding site & protein binding site;
3. DNA binding site complementary in shape and charge for specific DNA sequences;
4. Activators binding to enhancers + Repressors binding to silencers;
5. Protein binding site complementary in shape and charge proteins that they interact with;
 - a. Activator: recruit RNA polymerase / general transcription factors → help assembly of transcription initiation complex;
 - b. Repressors: sequester RNA polymerase / transcription factors /

block binding of transcription factors → prevent assembly of transcription initiation complex at the promoter;

- c. Activator: chromatin remodeling complexes & histone acetylase to tighten/loosen coiling of DNA around histones or increase accessibility of DNA;
 - d. Repressor: chromatin remodeling complexes & histone deacetylase to tighten/loosen coiling of DNA around histones or decrease accessibility of DNA;
6. Coordinated gene expression achieved combinations of activators & repressors which bind to enhancers & silencers which are associated with groups of genes with related functions;

(c) With reference to changes in protein structure, explain how the level of protein activity can be regulated post-translationally. [5]

- 1. Cleavage and/or covalent modification → change in conformation
- 2. Give 1 suitable example - glycosylation, disulfide bond formation, attachment of prosthetic groups etc. is required;
- 3. Form functional protein - newly synthesized proteins need to be modified for proper assembly / functioning;
- 4. Eg: phosphorylation/dephosphorylation → affect charge/conformational change on protein → may activate/inactivate (ORA) proteins
- 5. Eg. change folding to expose active sites of enzymes / allow for binding to another protein;
- 6. Regulate - control cellular activity / influence biological activity;
- 7. Addition of recognition tags for targeting E.g. Proteins are linked to ubiquitin;
- 8. → addition of recognition site for targeting to proteasome;
- 9. Degrade proteins allows control of protein activities OR prevent aberrant activities so that proteins will not stay too long in cytoplasm and still be active;

[Total: 20]

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